

Random events

The sample space first, the event second, the probability only third.

LESSON 91–93

3 × 40 MIN

ALGEBRA 10, §29

IGX 24.1

LESSON CLOCK

15'

30'

30'

30'

25'

5

Three kinds of event	15 min
The sample space	30 min
Tables and trees	30 min
Combining events	30 min
Exclusive and independent	25 min
Homework	5 min

LEARNING OBJECTIVES

- Distinguish certain, impossible and random events.
- List a sample space systematically, by table or by tree.
- Describe the union, intersection and complement of events.
- Recognise when two events are mutually exclusive or independent.

TEXTBOOK

National	pp. 305–318
Cambridge	Core & Extended, pp. 620–628
Workbook	Exercise 24.1

01

Explanation

Three kinds of event

A **trial** is anything with an uncertain result: a die thrown, a card drawn, a bulb tested. Its possible results are **outcomes**; a set of outcomes is an **event**.

KIND	HAPPENS	EXAMPLE (ONE DIE)
certain	always	“the score is under 7”
impossible	never	“the score is 8”
random	sometimes	“the score is even”

NOTATION USED THROUGHOUT

Events are $A, B, C, \dots$; the certain event is U (or Ω); the impossible event is $\emptyset$. The complement of A — “not A ” — is written A' or $\bar{A}$.

The sample space

Every probability question begins the same way: **list what can happen**. The list is the sample space, written U , and everything else is counted inside it.

TRIAL	SAMPLE SPACE	$ U $
one coin	$\{H, T\}$	2
one die	$\{1, 2, 3, 4, 5, 6\}$	6
two coins	$\{HH, HT, TH, TT\}$	4
two dice	ordered pairs (a, b)	36
a card from 52	the 52 cards	52
three coins	$\{HHH, \dots, TTT\}$	8

HT AND TH ARE DIFFERENT OUTCOMES

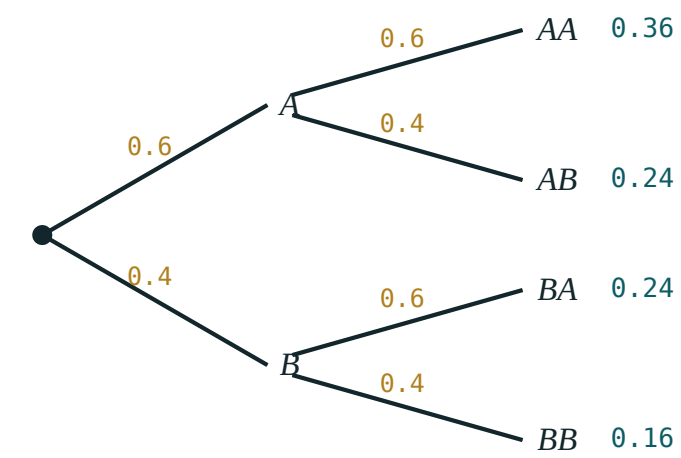
With two coins there are four equally likely outcomes, not three. “One head and one tail” happens in **two** of them, which is why its probability is $\frac{2}{4}$ and not $\frac{1}{3}$. Order the outcomes and the trap disappears.

Tables and trees

Two devices make the list systematic.

A table suits two trials with few outcomes each — two dice, for instance, give a 6×6 grid whose 36 cells are the sample space. Reading “the total is 7” off the leading anti-diagonal gives 6 cells at once.

A tree suits a sequence of stages, especially when the stages differ or when something is removed between them.



multiply along a branch · add down the ends

Each path through the tree is one outcome; the paths together are the sample space.

TWO RULES FOR TREES

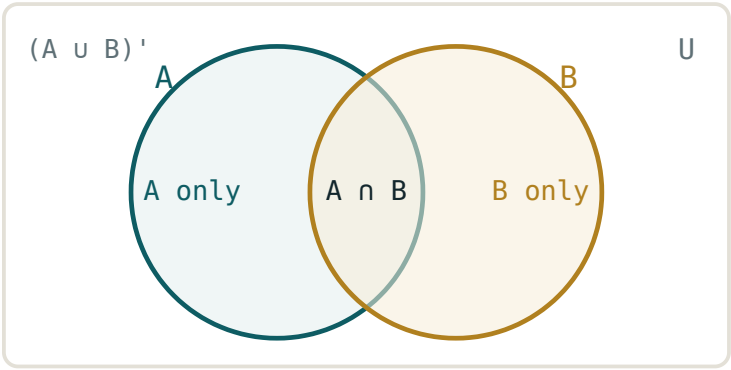
- **Along** a branch, multiply. The probabilities of one path multiply together.
- **Down** the ends, add. Several paths that all count as the event are added.

The probabilities on any one fork must total 1 — a check worth doing every time.

Combining events

Events are sets, so the set operations describe them.

IN WORDS	NOTATION	MEANS
A or B (or both)	$A \cup B$	the union
A and B	$A \cap B$	the intersection
not A	A'	the complement
A but not B	$A \cap B'$	the difference



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Two events, four regions — and every question in this block names one of them.

One die. Let $A = \text{“even”} = \{2, 4, 6\}$ and $B = \text{“greater than 3”} = \{4, 5, 6\}$. Then $A \cup B = \{2, 4, 5, 6\}$, $A \cap B = \{4, 6\}$, $A' = \{1, 3, 5\}$ and $A \cap B' = \{2\}$.

Exclusive and independent

Two words are constantly confused; they describe entirely different things.

	MUTUALLY EXCLUSIVE	INDEPENDENT
meaning	cannot both happen	one does not affect the other
test	$A \cap B = \emptyset$	$P(A \cap B) = P(A) \cdot P(B)$
Venn picture	circles apart	circles overlap, in proportion
example	“even” and “odd”	two throws of one die

EXCLUSIVE EVENTS ARE NEVER INDEPENDENT

If A and B cannot both happen, then knowing A occurred tells you B did **not** — that is the strongest possible dependence. The two words are almost opposites, not synonyms.

Replacement decides independence. Drawing two balls *with* replacement leaves the second draw untouched; drawing *without* replacement changes it. The phrase “without replacement” in a question is an instruction to change the second row of the tree.

02 Worked examples

EXAMPLE 1 Two dice are thrown. List the sample space and find how many outcomes give a total of 7.

- ① Ordered pairs (a, b) with $a, b \in \{1, \dots, 6\}$.
 $6 \times 6 = 36$ outcomes.

- ② $(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$

- ③ Six favourable outcomes.

ANSWER

$|U| = 36$; six outcomes give 7

EXAMPLE 2 A die is thrown. With $A = \text{“even”}$ and $B = \text{“prime”}$, list $A \cap B$ and $A \cup B$.

- ① $A = \{2, 4, 6\}$

- ② $B = \{2, 3, 5\}$
 1 is not prime.

- ③ $A \cap B = \{2\}$

- ④ $A \cup B = \{2, 3, 4, 5, 6\}$

ANSWER

$A \cap B = \{2\}, A \cup B = \{2, 3, 4, 5, 6\}$

EXAMPLE 3 A bag holds 3 red and 2 blue balls. Two are drawn without replacement. Draw the tree and list the outcomes.

① First fork: R with $\frac{3}{5}$, B with $\frac{2}{5}$.

② After R : $\frac{2}{4}$ and $\frac{2}{4}$.

Four balls left.

③ After B : $\frac{3}{4}$ and $\frac{1}{4}$.

④ Outcomes RR , RB , BR , BB .

ANSWER

Four outcomes, with the second fork depending on the first

EXAMPLE 4 Are “the first throw is a 6” and “the second throw is a 6” exclusive, independent, or neither?

① They can both happen — the throw $(6, 6)$.

Not exclusive.

② The first throw does not change the die.

③ $P(A \cap B) = \frac{1}{36} = \frac{1}{6} \times \frac{1}{6}$

Independent.

ANSWER

Independent, and not mutually exclusive

03 Interactive model

Throw two dice thirty times and tally the totals on the board; the shape of the answer appears before any formula.

Naming events correctly

The sample space for two coins has:

“The score on a die is 8” is:

A' means:

Two dice give how many outcomes?

12

21

36

6

Exclusive events satisfy:

$$A \cap B = \emptyset$$

$$A \cup B = U$$

$$P(A) = P(B)$$

$$A = B$$

Along a branch of a tree you:

add

multiply

subtract

divide

Drawing without replacement makes the draws:

independent

dependent

exclusive

certain

The probabilities on one fork total:

0

1

2

anything

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Random event	Tasodifiy hodisa	Случайное событие
Trial (experiment)	Sinov	Испытание
Outcome	Natija	Исход
Sample space	Elementar hodisalar fazosi	Пространство элементарных исходов
Certain event	Muqarrar hodisa	Достоверное событие
Impossible event	Mumkin bo'lmagan hodisa	Невозможное событие
Complementary event	Qarama-qarshi hodisa	Противоположное событие
Mutually exclusive	Birgalikda bo'lmagan	Несовместные
Independent events	Bog'liqsiz hodisalar	Независимые события
Tree diagram	Daraxt diagrammasi	Дерево исходов
Equally likely	Teng imkoniyatli	Равновозможные
Favourable outcome	Qulay natija	Благоприятный исход

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A sample space lists:

the favourable outcomes

every possible outcome

the events

the probabilities

With two coins, “one head” has:

one outcome

two outcomes

three

four

Mutually exclusive events are:

always independent

never independent (unless one is impossible)

the same thing

complementary

A tree diagram suits:

a single throw

a sequence of stages

a Venn question

a certain event

$A \cap B'$ means:

A or B

A but not B

neither

both

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Sample space for one coin

ANSWER $\{H, T\}$

2. Number of outcomes for two coins

ANSWER 4

3. Number of outcomes for two dice

ANSWER 36

4. Is “a die shows 7” certain, impossible or random?

ANSWER Impossible

5. Is “a die shows less than 7” certain, impossible or random?

ANSWER Certain

6. A = even on a die: list A'

ANSWER $\{1, 3, 5\}$

7. Number of outcomes for three coins

ANSWER 8

Medium · the standard question

7 PROBLEMS

1. $A = \text{even}, B = \text{prime on a die}: A \cap B$

ANSWER $\{2\}$

2. Same: $A \cup B$

ANSWER $\{2, 3, 4, 5, 6\}$

3. Two dice: how many outcomes give a total of 8?

ANSWER 5

4. Two dice: how many give a double?

ANSWER 6

5. Two coins: outcomes with exactly one head

ANSWER 2

6. A bag of 3 red, 2 blue: the second fork after a red, without replacement

ANSWER $\frac{2}{4}$ red, $\frac{2}{4}$ blue

7. Are “even” and “odd” on one die exclusive?

ANSWER Yes

Hard · stretch and reason

7 PROBLEMS

1. Two dice: how many outcomes have a total that is prime?

ANSWER 15

2. Three coins: how many outcomes have at least two heads?

ANSWER 4

3. A card from 52: are “red” and “king” exclusive? independent?

ANSWER Not exclusive; independent

4. Two dice: describe $A \cap B$ where A = “first is even”, B = “total is 5”

ANSWER $\{(2,3), (4,1)\}$

5. A bag of 4 white and 6 black: all outcomes of two draws without replacement

ANSWER WW, WB, BW, BB with changing forks

6. Explain why exclusive events with $P(A), P(B) > 0$ cannot be independent

ANSWER $P(A \cap B) = 0 \neq P(A)P(B)$

7. Two dice: how many outcomes give a total of at least 10?

ANSWER 6

07 Homework

Homework — 5 tasks

Every answer starts with the sample space written out or counted explicitly.

1. List the sample space for throwing a coin and a die together, and say how many outcomes it has.
2. For one die, let A = “multiple of 3” and B = “greater than 2”. List $A \cup B$, $A \cap B$, A' and $A \cap B'$.
3. Draw the tree for two draws from a bag of 5 green and 3 yellow counters, without replacement.
4. Give one pair of events that is exclusive but not independent, and one pair that is independent but not exclusive.

5. Make a 6×6 table for two dice and shade the cells where the total is 9.